

fMRI Foundation Model (V2) — Architecture (detailed)

A DINOv2 self-supervised foundation model on a multi-source fMRI corpus, evaluated by a leakage-free linear/MLP probe. Goal: pretrain a transferable encoder; downstream thesis target = ADNI progression.

0. Data pipeline (identical in training and probe)

Spatial. Every scan is downsampled offline to a fixed **(45, 54, 45)** volume.

Temporal TR harmonization. Native TR differs per dataset: HCP 0.72 s · ABIDE **per-site** (1.5–3.0 s) · OASIS 2.2 s · AOMIC 0.75/2.0 s · ADNI 3.0 s. To mix them we harmonize to a common **TARGET_TR = 0.72 s** and a fixed window **T = 270 frames** (= 194.4 s of brain activity): 1. native window length **win** = $\text{round}(270 \times 0.72 / \text{TR}_{\text{native}})$ frames — spans 194.4 s. 2. crop that window (on the mmap, so a 500 MB HCP scan never fully loads). 3. resample native→270 frames with **scipy.signal.resample_poly** (polyphase, anti-aliased FIR — the correct tool for band-limited BOLD).

Normalization. Per-frame spatial z-score: for each timepoint, subtract the mean over voxels and divide by the std over voxels → each 3D volume is zero-mean / unit-std. (*We do NOT z-score per-voxel over time — a candidate ablation.*)

Output of the pipeline: a **(270, 1, 45, 54, 45)** tensor.

1. Pretraining (self-supervised)

Objective — DINOv2. Self-distillation between a **student** and a **teacher** (the teacher = EMA of the student). Losses: - **DINO**: cross-entropy student teacher on the [CLS] token (with centering + sharpening), - **iBOT**: masked-token prediction, - **KoLeo**: feature-spreading regularizer.

Patchify — PatchEmbed3DPlus1D (turns the 4D scan into tokens): - **Spatial** Conv3d, kernel/stride **9** → $5 \times 6 \times 5 = 150$ spatial tokens. - **Temporal** Conv1d, kernel **10** → $270 / 10 = 27$ temporal tokens. - → token grid **27** × **150** = **4050 tokens**, each embedded to **384-d**.

Backbone. ViT-Small (embed 384), reg4, **initialized from the DINOv2 ViT-S/14 ImageNet checkpoint** (patch_embed + pos_embed dropped — shapes differ — and re-learned on fMRI).

Augmentation — masking-only. Crops = the **full volume** (2 global + N local, no spatial crop: a brain is a fixed anatomical structure); the only corruption is **per-token random masking** (MAE-style) applied in the collate for iBOT.

Batch composition — ProportionalBatchSampler. Fixed per-batch quota **HCP 4 / ABIDE 4 / OASIS 4 / ADNI 3 / AOMIC 1** = 16.

Corpus. 5 sources, **4627 scans**. Subject-level **holdout**: the 30% val+test subjects of HCP/ABIDE/OASIS/ADNI are **excluded from pretraining** (leakage-free probe); AOMIC is kept whole.

Freeze policy (base). Freeze DINOv2 blocks 0–8; train **patch_embed + blocks 9–11 + norm + heads** (fmri_plus_last_3).

Optimization. base_lr 3e-4, warmup 0.3 ep, **10 epochs**, patch_embed_lr_mult 0.2, effective batch 16 (micro-batch $2 \times \text{grad_accum}$ 8).

The 5 ablation runs (one factor each vs base): **base** (reference) · **fourier** (Fourier positional encoding) · **noblock2** (drop a patchify block) · **pool** (AvgPool downsampling vs strided conv) · **unfrozen** (all transformer layers trainable during SSL).

2. Probe (evaluation — encoder FROZEN)

Embedding extraction. Same spatial / TR / normalization pipeline as training. Per scan: slide 270-frame windows (stride = win/2), run each through the frozen **teacher**, take the [CLS] token per window, and **average** → one **384-d vector per scan** (cached to disk). *Difference vs training: training samples ONE random 270-window per step; the probe averages ALL windows for a stable, deterministic embedding.*

Probe head — Probe(nn.Module), PyTorch. hidden=() → single Linear (384→1) = logistic regression; hidden=(256,128) → MLP (Linear→ReLU→...→1). Trained with **Adam (lr 1e-3, wd 1e-4) + BCEWithLogitsLoss** (class-balanced pos_weight) + **backprop**, 200 epochs. Features standardized on the train split. **The encoder stays frozen** — only the head is trained.

Split — 70:30 leakage-free. In-corpus datasets use `subject_split.json` (the 30% test was held out of pretraining); external datasets (ADHD-200/COBRE/UCLA, never pretrained on) get a deterministic random 70:30.

Metrics. **AUROC** to rank the pretraining runs (threshold- and balance-independent); **Acc / F1** to match what the SOTA papers report.

Probed datasets. ADNI · ABIDE · HCP · OASIS (in-corpus, held-out 30%) + ADHD-200 · COBRE · UCLA (external, fully unseen).

3. Key points (honest)

- Strong **demographic** signal (HCP Sex AUROC 0.96); **clinical weak with a linear probe** → **fine-tuning** is the decisive next experiment.
- Same-dataset SOTA comparisons: **ADNI** (Brain-JEPA) and **ADHD-200** (NeuroSTORM). Brain-JEPA numbers are fine-tuning; ours are linear probe.
- Open questions: per-voxel temporal normalization (ablation), finishing the fine-tune, adding UCLA / HCP task-fMRI.